Candelas of the University of Texas at Austin, Gary Horowitz and Andrew Strominger of the University of California at Santa Barbara, and Edward Witten showed that a particular class of six-dimensional geometrical shapes can meet these conditions. They are known as Calabi-Yau spaces (or Calabi-Yau shapes) in honor of two mathematicians, Eugenio Calabi from the University of Pennsylvania and Shing-Tung Yau from Harvard University, whose research in a related context, but prior to string theory, plays a central role in understanding these spaces.

Although the mathematics describing Calabi-Yau spaces is intricate and subtle, we can get an idea of what they look like with a picture.8

In Figure 8.9 we show an example of a Calabi-Yau space.9 As you view this figure, you must bear in mind that the image has built-in limitations. We are trying to represent a six-dimensional shape on a two-dimensional piece of paper, and this introduces significant distortions. Nevertheless, the image does convey the rough idea of what a Calabi-Yau space looks like.10 The shape in Figure 8.9 is but one of many tens of thousands of examples of Calabi-Yau shapes that meet the stringent requirements for the extra dimensions that emerge from string theory. Although belonging to a club with tens of thousands of members might not sound very exclusive, you must compare it with the infinite number of shapes that are mathematically possible; by this measure Calabi-Yau spaces are rare indeed.

To put it all together, you should now imagine replacing each of the spheres in Figure 8.7—which represented two curled-up dimensions—with a Calabi-Yau space. That is, at every point in the three familiar extended dimensions, string theory claims that there are six hitherto unanticipated dimensions, tightly curled up into one of these rather complicated-looking shapes, as illustrated in Figure 8.10. These dimensions are an integral and ubiquitous part of the spatial fabric; they exist everywhere. For instance, if you sweep your hand in a large arc, you are moving not only through the three extended dimensions, but also through these curled-up dimensions. Of course, because the curled-up dimensions are so small, as you move your hand you circumnavigate them an enormous number of times, repeatedly returning to your starting point. Their tiny extent means that there is not much room for a large object like your hand to move—it all